v1.0Dark Matter Deep Dive Simulation

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: Dark Matter Benevolence Computational Framework **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** `v1.0Dark Matter Deep Dive Simulation V2.0.md`

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines' Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^7+ particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

---

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
WIMP direct detection	XENON1T 2018 era	LZ 2024: sigma_SI < 9.2×10^-48 cm^2 at m_χ=36 GeV — WIMP at weak-scale cross-section essentially ruled out	HIGH	V1.0 WIMP cross-section parameters now excluded by experiment
FDM mass constraints	No JWST data	JWST 2023: m_a > 10^-21 eV. FDM soliton core size updated. Granular halo texture more constrained	HIGH	V1.0 allows FDM masses that JWST rules out
WDM mass limit	m_WDM > 3.3 keV (V1.0)	Lyman-alpha 2024: m_WDM > 5.3 keV at 95% CL	MEDIUM	V1.0 allows WDM parameter values now excluded
Soliton breathing mode	Not covered	FDM soliton oscillates at f = 1/(4*r_c). V2.0 implements breathing mode in central galaxy region	LOW	Missing observable soliton dynamics in FDM simulation

---

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** FDM, WIMP
- **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
- **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
- **New interaction modes:** Triggered by new experimental discoveries

---

Unchanged Architecture

All foundational philosophy is **carried forward unchanged**: - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Dark Matter Deep Dive Simulation V1.0 archived | V2.0 is the active specification as of June 2026